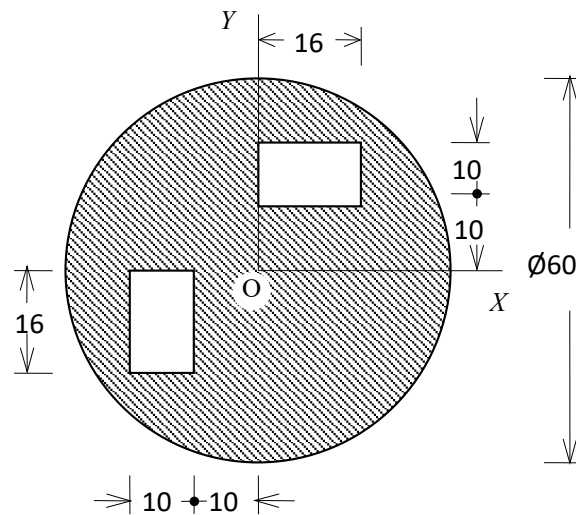


Additional exercises about geometry of areas

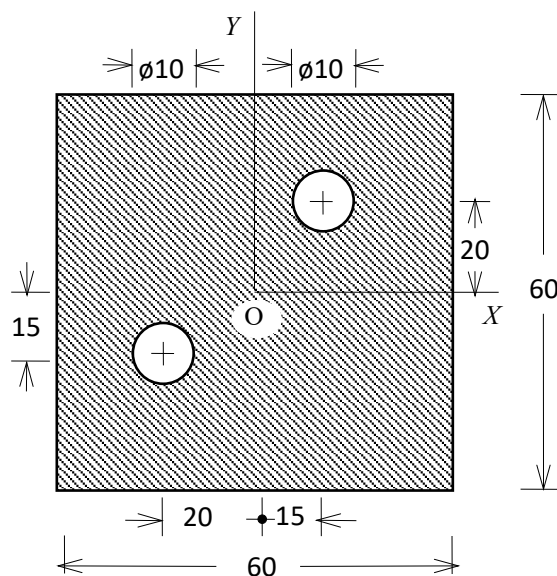
1. For the section shown below:

- a. find the coordinates of the centroid with respect to the (X, O, Y) reference system;
- b. compute the moments of the 2nd order with respect to a central reference system (x, G, y) ;
- c. compute the principal moments of inertia and the orientation of the principal reference system $(1, G, 2)$ with respect to the central reference system (x, G, y) .



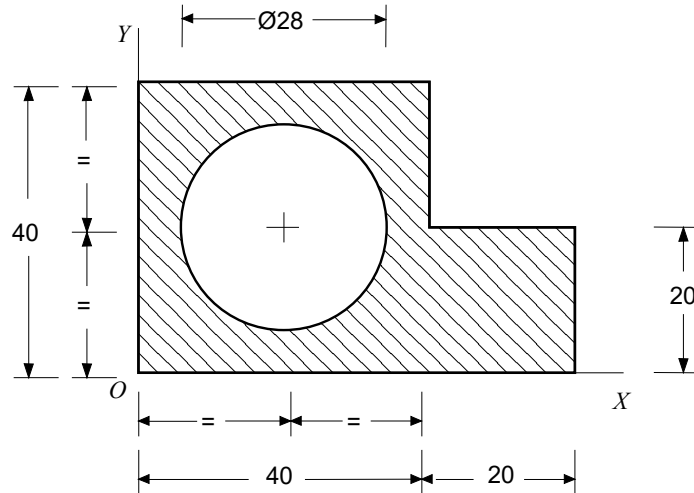
2. For the section shown below:

- a. find the coordinates of the centroid with respect to the (X, O, Y) reference system;
- b. compute the moments of the 2nd order with respect to a central reference system (x, G, y) ;
- c. compute the principal moments of inertia and the orientation of the principal reference system $(1, G, 2)$ with respect to the central reference system (x, G, y) .



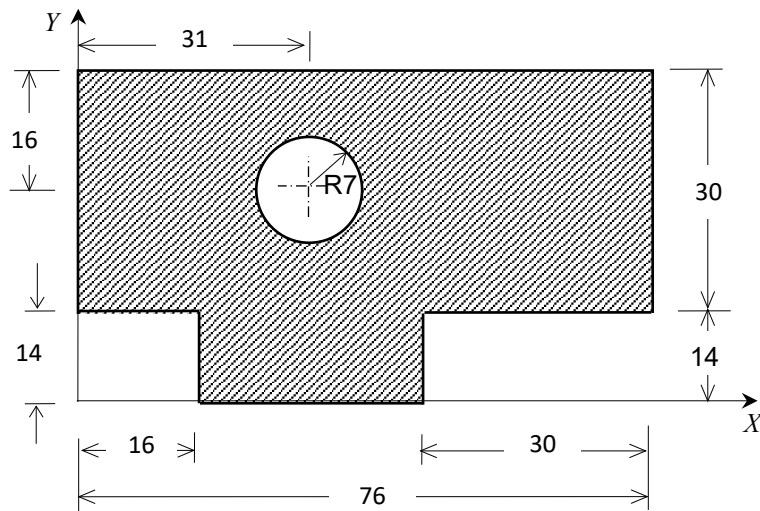
3. For the section shown below:

- find the coordinates of the centroid with respect to the (X, O, Y) reference system;
- compute the moments of the 2nd order with respect to a central reference system (x, G, y) ;
- compute the principal moments of inertia and the orientation of the principal reference system $(1, G, 2)$ with respect to the central reference system (x, G, y) .



4. For the section shown below:

- find the coordinates of the centroid with respect to the (X, O, Y) reference system;
- compute the moments of the 2nd order with respect to a central reference system (x, G, y) ;
- compute the principal moments of inertia and the orientation of the principal reference system $(1, G, 2)$ with respect to the central reference system (x, G, y) .



5. For the section shown below:

- find the coordinates of the centroid with respect to the (X,O,Y) reference system;
- compute the moments of the 2nd order with respect to a central reference system (x,G,y) ;
- compute the principal moments of inertia and the orientation of the principal reference system $(1,G,2)$ with respect to the central reference system (x,G,y) .

